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CALIBRATION PROCEDURE

Volumetric Glassware & Instruments — per EURAMET Calibration Guide No. 19

1. Scope & Application

This procedure applies to the gravimetric calibration of volumetric instruments and glassware — including pipettes, burettes, volumetric flasks, and standard capacity measures — per EURAMET Calibration Guide No. 19, "Guidelines on the Determination of Uncertainty in Gravimetric Volume Calibration", and EURAMET Calibration Guide No. 21 for standard capacity measures.

2. Reference Standards & Traceability

STANDARD	DESCRIPTION	TRACEABILITY
Analytical Balance	Resolution and capacity appropriate to the volume under test (gravimetric method)	Calibrated per QSL-CAL-PROC-001 (NAWI), ≤ 12 months
Reference Thermometer	Calibration liquid (distilled water) temperature measurement	Accredited lab, ≤ 24 months
Barometer & Hygrometer	Ambient pressure and humidity for air density determination	Accredited lab, ≤ 12 months

3. Environmental Conditions

PARAMETER	REQUIREMENT
Ambient Pressure	940–1080 hPa
Ambient Temperature	18–30°C, recorded throughout the test
Calibration Liquid	Distilled or deionised water, equilibrated to ambient temperature before use

4. Procedure

- Pre-Test Inspection.** Inspect the instrument for cracks, chips, contamination, or (for capacity measures) damage to the discharge valve and graduated scale; clean as required.
- Levelling.** Level the instrument where applicable (e.g. proving tanks) before commencing.
- Tare.** Weigh the empty (dry) receiving vessel on the analytical balance and record the tare mass.
- Fill / Deliver.** Fill or deliver the nominal volume of calibration liquid per the instrument's designated method (e.g. delivery for pipettes and burettes, observing the specified drain/dripping time; filling to the graduation line for flasks).
- Weigh.** Weigh the receiving vessel with the delivered liquid and calculate the net mass of liquid delivered.
- Repeat Determinations.** Repeat steps 3–5 a minimum of 5 times (or 10 for higher-precision requirements) to establish repeatability.
- Convert Mass to Volume.** Convert the mean net mass to volume using the density of the calibration liquid at the measured temperature, corrected for air buoyancy.

5. Uncertainty & Calculation

The combined standard uncertainty includes: the calibration uncertainty of the balance; the repeatability of the gravimetric determinations; the uncertainty of the calibration liquid density (as a function of measured temperature); the uncertainty of the air density correction; and the meniscus-reading uncertainty for glassware with a graduated scale. These are combined per the GUM method to give the expanded uncertainty ($k=2$) of the determined volume.

6. Acceptance Criteria

The instrument passes calibration where the deviation of the determined volume from the nominal volume, together with its uncertainty, falls within the manufacturer's stated tolerance class (e.g. Class A or Class B for glassware) or the client's stated tolerance requirement.

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